Practice Quiz

Gemini typeset problems dictated to it and chose the specific matrix A. The file was then moderately edited.

Show all your work clearly. Write precisely. **Justify everything.** No credit for unsupported answers.

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1. Consider the matrix $A \in \mathbb{R}^{3\times 4}$ given by:

$$A = \begin{pmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 1 & 5 \\ 3 & 6 & 0 & 9 \end{pmatrix}$$

(a) Perform elementary row operations to transform A into a **Row Echelon Form (REF)**. Clearly specify the row operation performed at every step.

(b) Further reduce the matrix to a **Reduced Row Echelon Form (RREF)**.

2. Consider the matrix A defined in Problem 1.

(a) Find the complete solution set of the homogeneous system $A\mathbf{x} = \mathbf{0}$, where $\mathbf{x} = (x_1, x_2, x_3, x_4)^T \in \mathbb{R}^4$.

(b) Find a basis for this solution set and justify why your answer is a basis.

3. Again consider the matrix A in problem 1. Find an explicit column vector $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \in \mathbb{R}^3$ such that the linear system $A\mathbf{x} = \mathbf{b}$ is **inconsistent** (i.e., has no solutions). Justify.

4. Let M be an arbitrary $r \times c$ matrix with entries in $\mathbb{R}$. Prove that the **pivot positions** in any row echelon form of M are uniquely determined by M, that is, these positions are independent of the particular sequence of elementary row operations used to reduce M to an REF. (Note: there are potentially several ways to argue this. See if you can find a short "proof by language" based only on the method to solve a linear system by row reduction. The key is to find carefully chosen wording to isolate a relevant aspect of this procedure.)